

一、手工创建数据库

图 1. 数据库创建成功截图:

```
Total System Global Area 1795159104 bytes
Fixed Size                  8897600 bytes
Variable Size               419430400 bytes
Database Buffers            1358954496 bytes
Redo Buffers                 7876608 bytes
create database seujw03
datafile '/u01/app/oracle/oradata/seujw03/system01.dbf' size 400m
sysaux datafile '/u01/app/oracle/oradata/seujw03/sysaux01.dbf' size 400m
default tablespace users datafile '/u01/app/oracle/oradata/seujw03/users01.dbf'
size 100m
default temporary tablespace temp tempfile '/u01/app/oracle/oradata/seujw03/tem
p01.dbf' size 20m
undo tablespace undotbs1 datafile '/u01/app/oracle/oradata/seujw03/undotbs01.db
f' size 50m
logfile
group 1 '/u01/app/oracle/oradata/seujw03/redo01.log' size 10m,
group 2 '/u01/app/oracle/oradata/seujw03/redo02.log' size 10m,
14:18:37 10 group 3 '/u01/app/oracle/oradata/seujw03/redo03.log' size 10m;

Database created.

14:18:58 SYS@seujw03>
```

图 2. 创建数据字典视图截图

```
Synonym created.

Grant succeeded.

PL/SQL procedure successfully completed.

Grant succeeded.

Synonym created.

Session altered.

Session altered.

PL/SQL procedure successfully completed.

TIMESTAMP
-----
COMP_TIMESTAMP CATALOG    2026-06-18 14:21:37

Session altered.

Session altered.

14:21:37 SYS@seujw03>
```

```

14:44:52 SYS@seuj lw03>Rem
14:44:52 SYS@seuj lw03>Rem
NAME
      sqlsessend.sql - SQL session end
14:44:52 SYS@seuj lw03>Rem
DESCRIPTION
      Any commands which should be run at the end of all oracle
      supplied scripts.
14:44:52 SYS@seuj lw03>Rem
NOTES
      See sqlsessstart.sql for the corresponding start script.
14:44:52 SYS@seuj lw03>Rem
BEGIN SQL_FILE_METADATA
SQL_SOURCE_FILE: rdbms/admin/sqlsessend.sql
SQL_SHIPPED_FILE: rdbms/admin/sqlsessend.sql
SQL_PHASE: MISC
SQL_STARTUP_MODE: NORMAL
SQL_IGNOREABLE_ERRORS: NONE
END SQL_FILE_METADATA
14:44:52 SYS@seuj lw03>Rem
MODIFIED      (MM/DD/YY)
surman        05/04/18 - 27464252: Update SQL_PHASE
surman        03/08/13 - 16462837: Common start and end scripts
surman        03/08/13 - Created
14:44:52 SYS@seuj lw03>Rem
14:44:52 SYS@seuj lw03>alter session set "_ORACLE_SCRIPT" = false;
Session altered.
14:44:52 SYS@seuj lw03>
14:44:52 SYS@seuj lw03>
14:44:52 SYS@seuj lw03>
14:44:52 SYS@seuj lw03>Rem *****
14:44:52 SYS@seuj lw03>Rem END catproc.sql
14:44:52 SYS@seuj lw03>Rem *****
14:44:52 SYS@seuj lw03>

```

```

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SQL_STARTUP_MODE: NORMAL
SQL_IGNOREABLE_ERRORS: NONE
END SQL_FILE_METADATA
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Session altered.
14:44:52 SYS@seuj lw03>
14:44:52 SYS@seuj lw03>
14:44:52 SYS@seuj lw03>Rem *****
14:44:52 SYS@seuj lw03>Rem END catproc.sql
14:44:52 SYS@seuj lw03>Rem *****
14:44:52 SYS@seuj lw03>

```

图 4. 创建 scott 方案截图

```
Comment created.  
  
Comment created.  
  
Comment created.  
  
Comment created.  
  
Comment created.  
  
Comment created.  
  
Comment created.  
  
Commit complete.  
  
PL/SQL procedure successfully completed.  
14:47:08 SYS@seujlw03>
```

图 5. 创建 hr 方案截图

```
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

14:45:50 SYS@seujlw03>@?/rdbms/admin/utlsamp1.sql
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@ocpjlw ~]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Thu Jun 18 14:46:17 2026
Version 19.3.0.0.0

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Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

14:46:17 SYS@seujlw03>@/u01/app/oracle/product/19.3.0/dbhome_1/demo/schema/human_resources/hr_main.sql

specify password for HR as parameter 1:
Enter value for 1: hr

specify default tablespace for HR as parameter 2:
Enter value for 2: users

specify temporary tablespace for HR as parameter 3:
Enter value for 3: temp

specify log path as parameter 4:
Enter value for 4: /u01/log

PL/SQL procedure successfully completed.
```

二、SQL 语句

查询 hr.employees 表，显示 3 个字段：last_name, years, months，即员工姓名，工作了多少年零多少个月

条件：工作时间时长在第 6 到 20 长，按照 years 和 months 降序排序

输出示例：

```
LAST_NAME YEARS MONTHS
DeHean 24 4
```

图 6. SQL 语句截图

```
SELECT
    last_name,
    years,
    months
FROM (
    SELECT
        last_name,
        TRUNC(TRUNC(MONTHS_BETWEEN(CURRENT_DATE, hire_date)) / 12) AS years,
        MOD(TRUNC(MONTHS_BETWEEN(CURRENT_DATE, hire_date)), 12) AS months,
        ROW_NUMBER() OVER (
            ORDER BY MONTHS_BETWEEN(CURRENT_DATE, hire_date) DESC, last_name
        ) AS rn
    FROM hr.employees
)
WHERE rn BETWEEN 6 AND 20
14:48:52 16 ORDER BY years DESC, months DESC;
```

图 7. SQL 查询结果截图

LAST_NAME	YEARS	MONTHS
Faviet	23	10
Greenberg	23	10
Raphaely	23	6
Kaufling	23	1
Khoo	23	1
King	23	0
Ladwig	22	11
Whalen	22	9
Rajs	22	8
Sarchand	22	4
King	22	4
Bell	22	4
Hartstein	22	4
Sully	22	3
Abel	22	1
15 rows selected.		

三、备份和恢复

图 8. rman 全库备份过程截图

```

RMAN> backup database plus archivelog;

Starting backup at 18-JUN-26
current log archived
using channel ORA_DISK_1
channel ORA_DISK_1: starting archived log backup set
channel ORA_DISK_1: specifying archived log(s) in backup set
input archived log thread=1 sequence=175 RECID=1 STAMP=1236265031
input archived log thread=1 sequence=176 RECID=2 STAMP=1236265056
channel ORA_DISK_1: starting piece 1 at 18-JUN-26
channel ORA_DISK_1: finished piece 1 at 18-JUN-26
piece handle=/u01/demo/bak/SEUJLW03_014qvq30_20260618 tag=TAG20260618T145736 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01
Finished backup at 18-JUN-26

Starting backup at 18-JUN-26
using channel ORA_DISK_1
channel ORA_DISK_1: starting full datafile backup set
channel ORA_DISK_1: specifying datafile(s) in backup set
input datafile file number=00001 name=/u01/app/oracle/oradata/seujlw03/system01.dbf
input datafile file number=00002 name=/u01/app/oracle/oradata/seujlw03/sysaux01.dbf
input datafile file number=00004 name=/u01/app/oracle/oradata/seujlw03/users01.dbf
input datafile file number=00003 name=/u01/app/oracle/oradata/seujlw03/undotbs01.dbf
channel ORA_DISK_1: starting piece 1 at 18-JUN-26
channel ORA_DISK_1: finished piece 1 at 18-JUN-26
piece handle=/u01/demo/bak/SEUJLW03_024qvq31_20260618 tag=TAG20260618T145737 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:03
Finished backup at 18-JUN-26

Starting backup at 18-JUN-26
current log archived
using channel ORA_DISK_1
channel ORA_DISK_1: starting archived log backup set
channel ORA_DISK_1: specifying archived log(s) in backup set
input archived log thread=1 sequence=177 RECID=3 STAMP=1236265061
channel ORA_DISK_1: starting piece 1 at 18-JUN-26
channel ORA_DISK_1: finished piece 1 at 18-JUN-26
piece handle=/u01/demo/bak/SEUJLW03_034qvq35_20260618 tag=TAG20260618T145741 comment=NONE
channel ORA_DISK_1: backup set complete, elapsed time: 00:00:01
Finished backup at 18-JUN-26

Starting Control File and SPFILE Autobackup at 18-JUN-26
piece handle=/u01/demo/bak/auto/c-4114700926-20260618-00 comment=NONE
Finished Control File and SPFILE Autobackup at 18-JUN-26

RMAN> list backup;
```

图 9. rman 全库备份结果截图

```

RMAN> list backup;

List of Backup Sets
=====

BS Key   Size                Device Type Elapsed Time Completion Time
-----
1        4.57M            DISK         00:00:00      18-JUN-26
        BP Key: 1      Status: AVAILABLE Compressed: NO Tag: TAG20260618T145736
        Piece Name: /u01/demo/bak/SEUJLW03_014qvq30_20260618

        List of Archived Logs in backup set 1
        Thrd Seq      Low SCN    Low Time   Next SCN   Next Time
        ---
1        175      579393    18-JUN-26 584314     18-JUN-26
1        176      584314    18-JUN-26 584370     18-JUN-26

BS Key   Type LV Size                Device Type Elapsed Time Completion Time
-----
2        Full 486.05M            DISK         00:00:02      18-JUN-26
        BP Key: 2      Status: AVAILABLE Compressed: NO Tag: TAG20260618T145737
        Piece Name: /u01/demo/bak/SEUJLW03_024qvq31_20260618

        List of Datafiles in backup set 2
        File LV Type Ckp SCN      Ckp Time   Abs Fuz SCN Sparse Name
        ---
1        Full 584382    18-JUN-26                NO /u01/app/oracle/oradata/seujlw03/system01.dbf
2        Full 584382    18-JUN-26                NO /u01/app/oracle/oradata/seujlw03/sysaux01.dbf
3        Full 584382    18-JUN-26                NO /u01/app/oracle/oradata/seujlw03/undotbs01.dbf
4        Full 584382    18-JUN-26                NO /u01/app/oracle/oradata/seujlw03/users01.dbf

BS Key   Size                Device Type Elapsed Time Completion Time
-----
3        8.00K            DISK         00:00:00      18-JUN-26
        BP Key: 3      Status: AVAILABLE Compressed: NO Tag: TAG20260618T145741
        Piece Name: /u01/demo/bak/SEUJLW03_034qvq35_20260618

        List of Archived Logs in backup set 3
        Thrd Seq      Low SCN    Low Time   Next SCN   Next Time
        ---
1        177      584370    18-JUN-26 584389     18-JUN-26

BS Key   Type LV Size                Device Type Elapsed Time Completion Time
-----
4        Full 8.33M            DISK         00:00:00      18-JUN-26
        BP Key: 4      Status: AVAILABLE Compressed: NO Tag: TAG20260618T145742
        Piece Name: /u01/demo/bak/auto/c-4114700926-20260618-00
        SPFILE Included: Modification time: 18-JUN-26
        SPFILE db_unique_name: SEUJLW03
        Control File Included: Ckp SCN: 584402      Ckp time: 18-JUN-26
RMAN>

```

图 10. 删除 scott.emp 表语句和结果截图

```
15:02:35 SYS@seujlw03>conn scott/tiger
Connected.
15:03:03 SCOTT@seujlw03>drop table scott.emp purge;

Table dropped.

15:03:34 SCOTT@seujlw03>commit
15:06:10      2  ;

Commit complete.

15:06:12 SCOTT@seujlw03>select * from scott.emp;
select * from scott.emp
                        *
ERROR at line 1:
ORA-00942: table or view does not exist
```

图 11. 利用 logmnr 日志挖掘器查询出删除 scott.emp 的具体时间截图

```
15:09:02 SYS@seujlw03>select TIMESTAMP, USERNAME, SQL_REDO from v$logmnr_contents where SEG_NAME='EMP';

TIMESTAMP
-----
USERNAME
-----
SQL_REDO
-----
2026-06-18 15:03:34
SCOTT
drop table scott.emp purge;
```


图 12. scott.emp 表恢复结果截图

```

RMAN> alter database open resetlogs;
Statement processed
RMAN> select * from scott.emp;

```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL
7369	SMITH	CLERK	7902	1980-12-17 00:00:00	800
7499	ALLEN	SALESMAN	7698	1981-02-20 00:00:00	1600
7521	WARD	SALESMAN	7698	1981-02-22 00:00:00	1250
7566	JONES	MANAGER	7839	1981-04-02 00:00:00	2975
7654	MARTIN	SALESMAN	7698	1981-09-28 00:00:00	1250
7698	BLAKE	MANAGER	7839	1981-05-01 00:00:00	2850
7782	CLARK	MANAGER	7839	1981-06-09 00:00:00	2450
7788	SCOTT	ANALYST	7566	1987-04-19 00:00:00	3000
7839	KING	PRESIDENT		1981-11-17 00:00:00	5000
7844	TURNER	SALESMAN	7698	1981-09-08 00:00:00	1500
7876	ADAMS	CLERK	7788	1987-05-23 00:00:00	1100
7900	JAMES	CLERK	7698	1981-12-03 00:00:00	950
7902	FORD	ANALYST	7566	1981-12-03 00:00:00	3000
7934	MILLER	CLERK	7782	1982-01-23 00:00:00	1300

```

14 rows selected
RMAN>

```

四、闪回事务查询

图 13. 添加数据库补充日志截图

```
15:31:13 SYS@seujlw03>select SUPPLEMENTAL_LOG_DATA_MIN from v$database;
SUPPLEMENTAL_LOG
-----
NO

15:31:17 SYS@seujlw03>alter database add SUPPLEMENTAL LOG DATA;
Database altered.

15:31:25 SYS@seujlw03>select SUPPLEMENTAL_LOG_DATA_MIN from v$database;
SUPPLEMENTAL_LOG
-----
YES
```

图 14. 通过版本查询得到事务号截图

```
15:33:52 SYS@seujlw03>select versions_xid,empno,ename,versions_operation from scott.my_emp versions between scn minvalue and maxvalue
;

```

VERSIONS_XID	EMPNO	ENAME	VE
05002100A4010000	7844	TURNER	D
05002100A4010000	7900	JAMES	D
	7369	SMITH	
	7499	ALLEN	
	7521	WARD	
	7566	JONES	
	7654	MARTIN	
	7698	BLAKE	
	7782	CLARK	
	7788	SCOTT	
	7839	KING	
	7844	TURNER	
	7876	ADAMS	
	7900	JAMES	
	7902	FORD	
	7934	MILLER	

```
16 rows selected.
```

图 15. 进行事务查询得到 UNDO_SQL 截图

```
15:35:52 SYS@seujlw03>select undo_sql from FLASHBACK_TRANSACTION_QUERY where xid='05002100A4010000';
UNDO_SQL
-----
insert into "SCOTT"."MY_EMP"("EMPNO","ENAME","JOB","MGR","HIREDATE","SAL","COMM","DEPTNO") values ('7844','TURNER','SALE
SMAN','7698',TO_DATE('1981-09-08 00:00:00','yyyymm-dd hh24:mi:ss'),'1500','0','30');

insert into "SCOTT"."MY_EMP"("EMPNO","ENAME","JOB","MGR","HIREDATE","SAL","COMM","DEPTNO") values ('7900','JAMES','CLERK
','7698',TO_DATE('1981-12-03 00:00:00','yyyymm-dd hh24:mi:ss'),'950',NULL,'30');
```